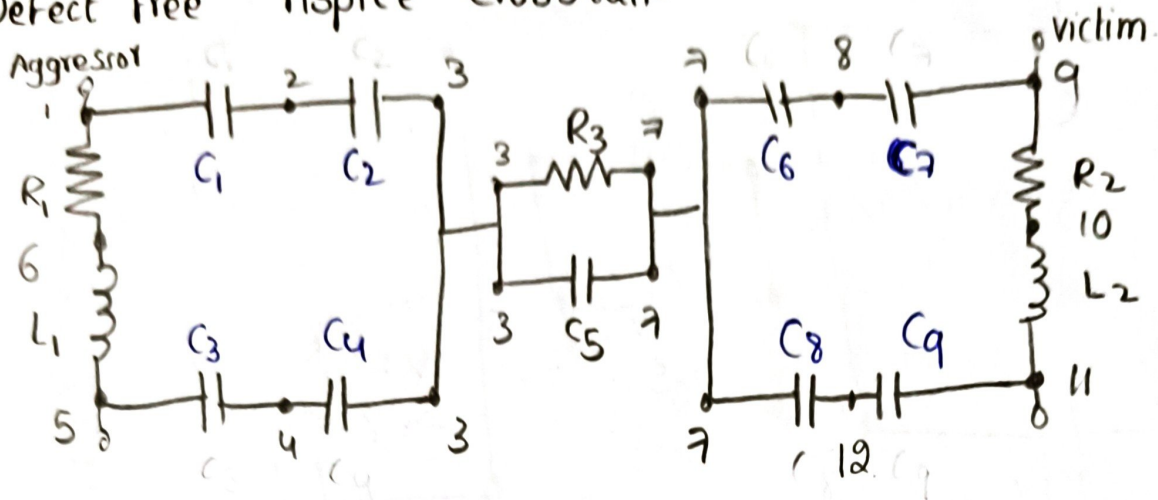


17/06/26 Day 22 Kamineni Dhana Sri Keerthi

Δ Defect free Hspice Crosstalk.



Code

```

*temp25
* Supply *
Vagg 1 0 pulse (0 1.8 0s 100p 100p 5n 10n)
Vg01 5 0 DC 0
Vg02 11 0 DC 0

```

Network Day 20th Notes. Reference code network.

operation

- option post
- tran ip 100n.

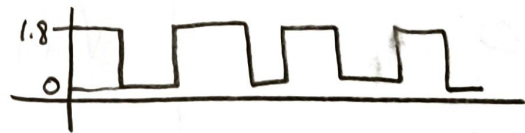
→ adding
 C1 5 0 1f
 C12 11 0 1.5f.

measure

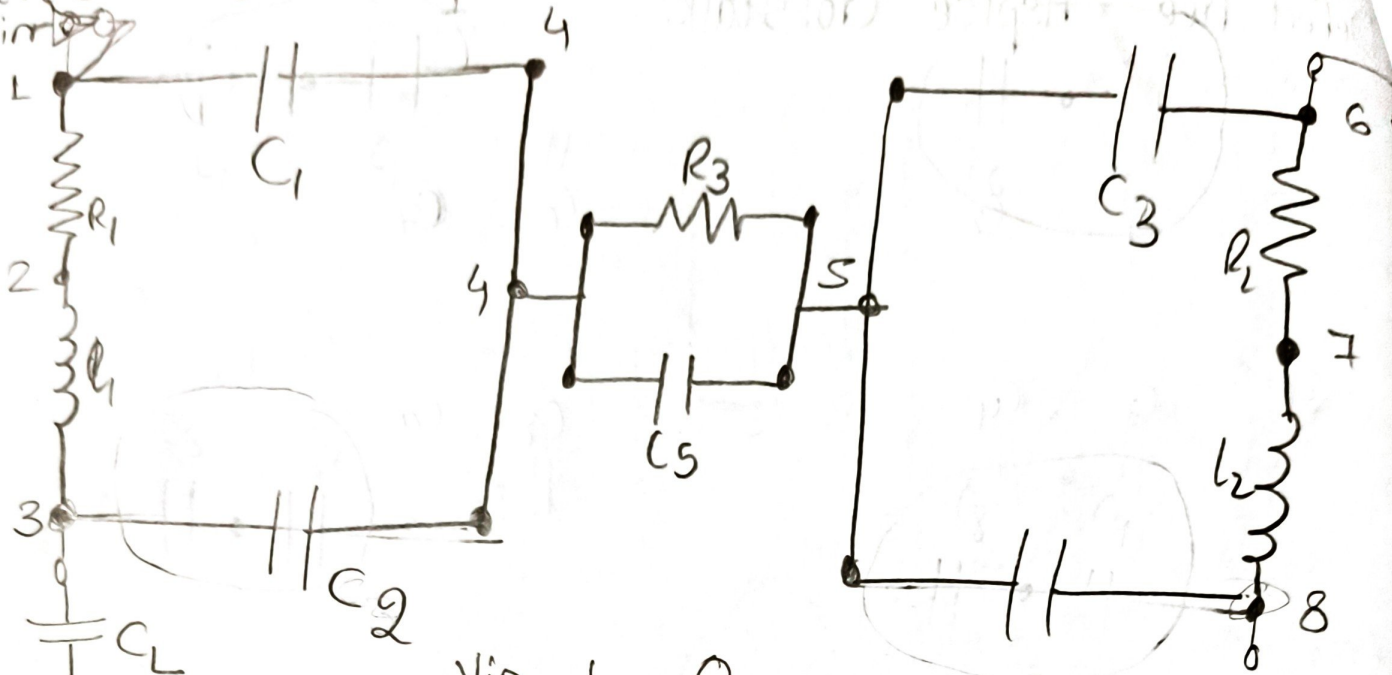
- PROBE V(1) V(3) V(7) V(9) V(10)
- end

Meaning

or V(1) Signal node (aggressor input)

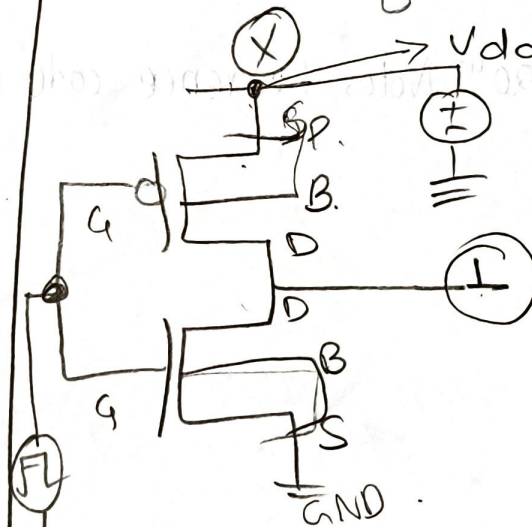


DEFECT FREE



V_{in} 1 0
 V_{L1} 6 0 DC 0 C_4

$V(3)$ V_{in}
 $V(8)$ 0 V_{dd} $\rightarrow$ peak noise
 0 0.8V



V_{in} 4 0 pulse

** Driver
 D G S B
 M_P 1 4 X X
 M_n 1 4 0 0

PMOS $W=8.1\mu$
 NMOS $W=4\mu$ $L=1\mu$

R_1	1	2	82.8m
R_2	6	7	82.8m
R_3	4	5	354 μ
L_1	2	3	2.097p
L_2	7	8	2.097p
C_1	1	4	11.55f
C_2	3	4	11.55f
C_3	5	6	11.55f
C_4	5	8	11.55f
C_5	4	5	3.93f
C_{L1}	3	0	7.91f
C_{L2}	8	0	3.91f

23.11f

measure Tot 1 $\rightarrow$ measure peak to peak.
measure delay.

3.1u = noise.

delay $V(1)$ $V(3)$ $\rightarrow$ 0.5uF, falling edge.

$$\text{Propagation delay} = \frac{t_r + t_f}{2}$$

$$t_r = 0.45\mu\text{F}$$

$$t_f = 0.52\mu\text{F}$$

$$\frac{0.45 + 0.52}{2} = \frac{0.97}{2} = 0.485$$

Inphase (cross talk delay)

$$T = 20.7\mu\text{F}$$

$$f = 12.3$$

$$\frac{20.7 + 12.3}{2} = 16.5\mu\text{F}$$

Code

Measurement
peak to peak.

delay $\sim$ risetime

~~rise~~ ~~same~~ ~~V(1)~~ ~~V(3)~~

Signal $V(1)$

Ref sig $V(2)$.

Signal $S0$

Ref sig $S0$.

Edge type

rise time $\rightarrow$ rise, same

fall time $\rightarrow$ fall same

inphase $V(1)$ $V(8)$.

measure - delay $V1a$ 0 0.8
 $V1v$ 0 0.8.

$V(1)$
 $V(8)$

50
10

rise
same.

outphase, $V1a$ 0. 0.8

$V1v$ 0.8 0.

$$V = 90.5\mu\text{F}$$

$$F = 11\mu\text{sp.}$$

$$= 117.75 \text{ outphase crosstalk delay}$$

$$\frac{t_r + t_f}{2} = \text{propagation delay}$$

level me.

$$\text{noise @ peak to peak}$$

$$\frac{2}{2}$$

pdelay
noise

cross in
out